Where's Waldo? Query–language Preference under Cross–lingual Knowledge Disparities

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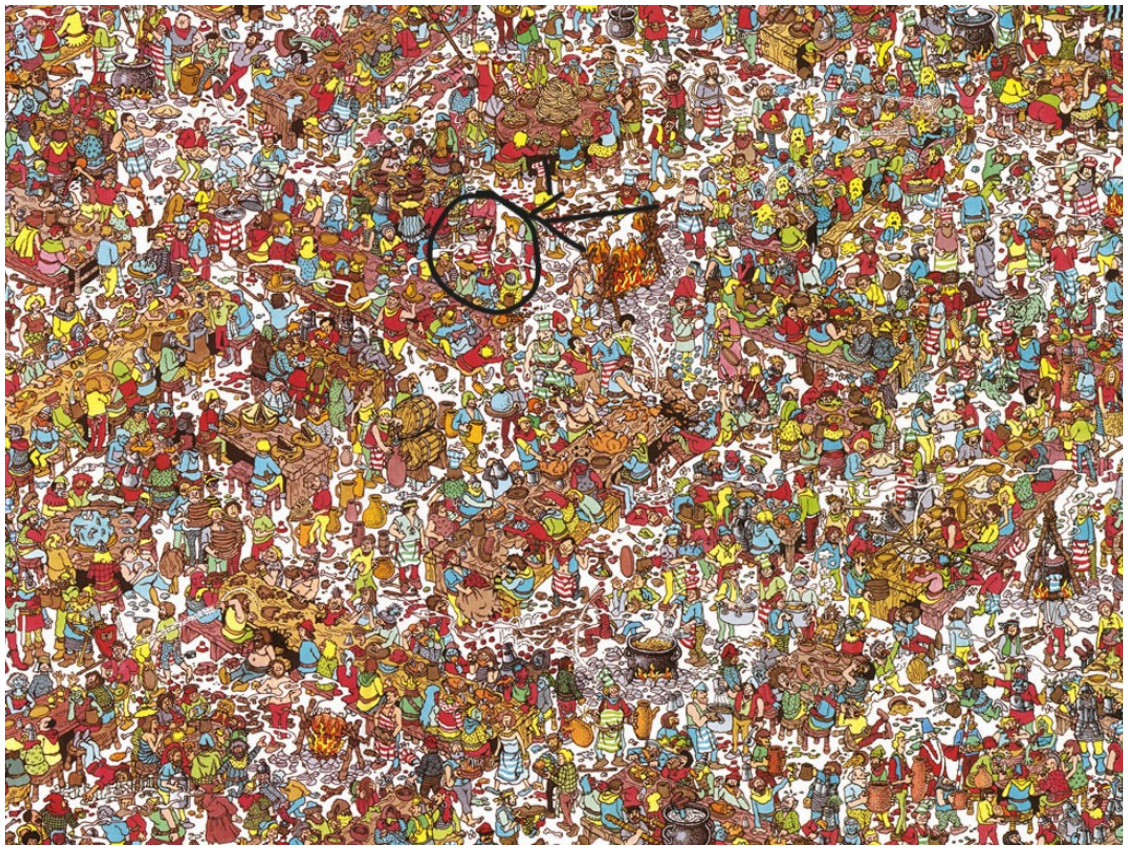
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The paper

What's Where's Waldo?



Where's Waldo? illustration: Martin Handford · supplied example

Find one distinctive person in a crowded scene.

Red-and-white stripes. Glasses. A matching hat.

The paper's version: find the relevant fact across languages.

WALDO = WikipediA Lost in Disparity of knOwledge

The paper

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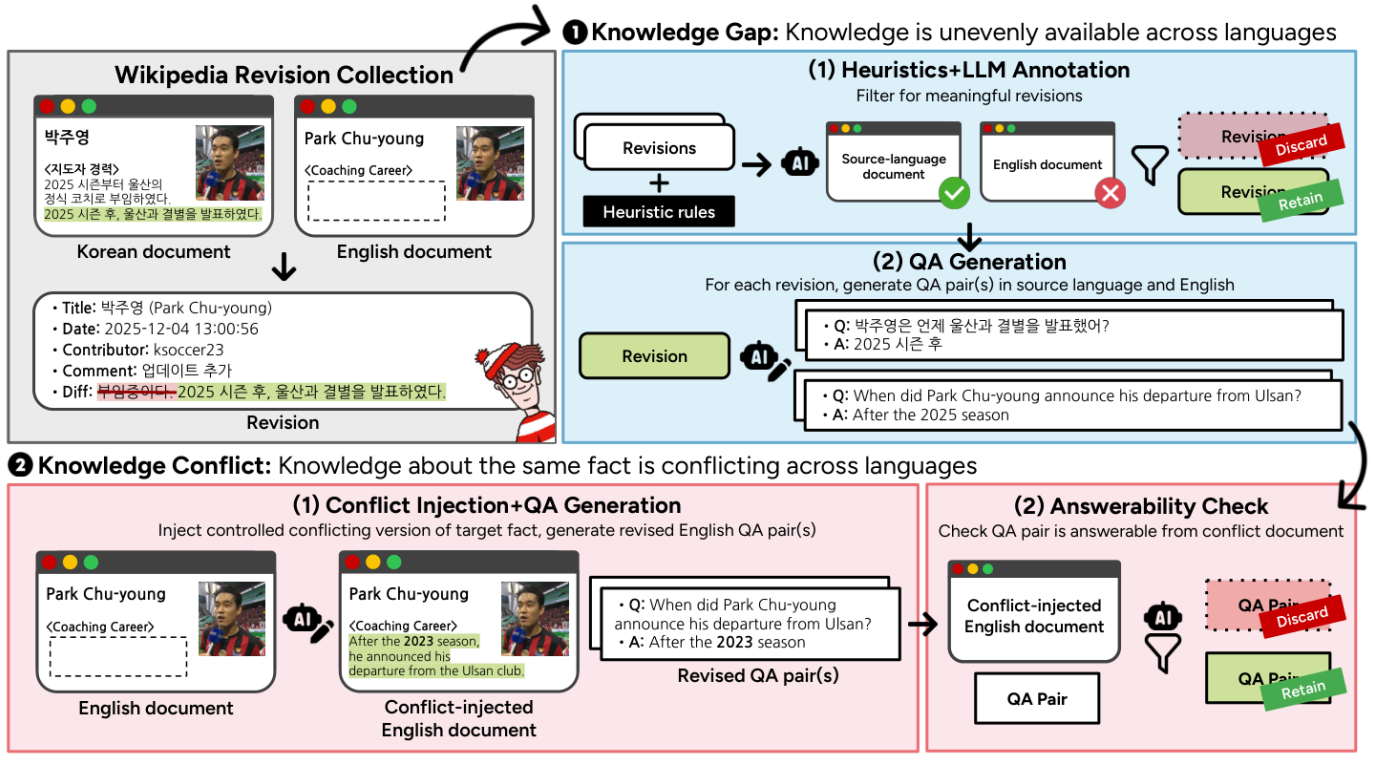
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Work conducted during Dayeon Ki's internship at Microsoft.

Design

Knowledge gaps and conflicts



Ki et al. (2026), Figure 1

≈12K QA pairs · 5 source languages + English · 8 models

Gap

Korean: after 2025.
English: fact absent.

Conflict

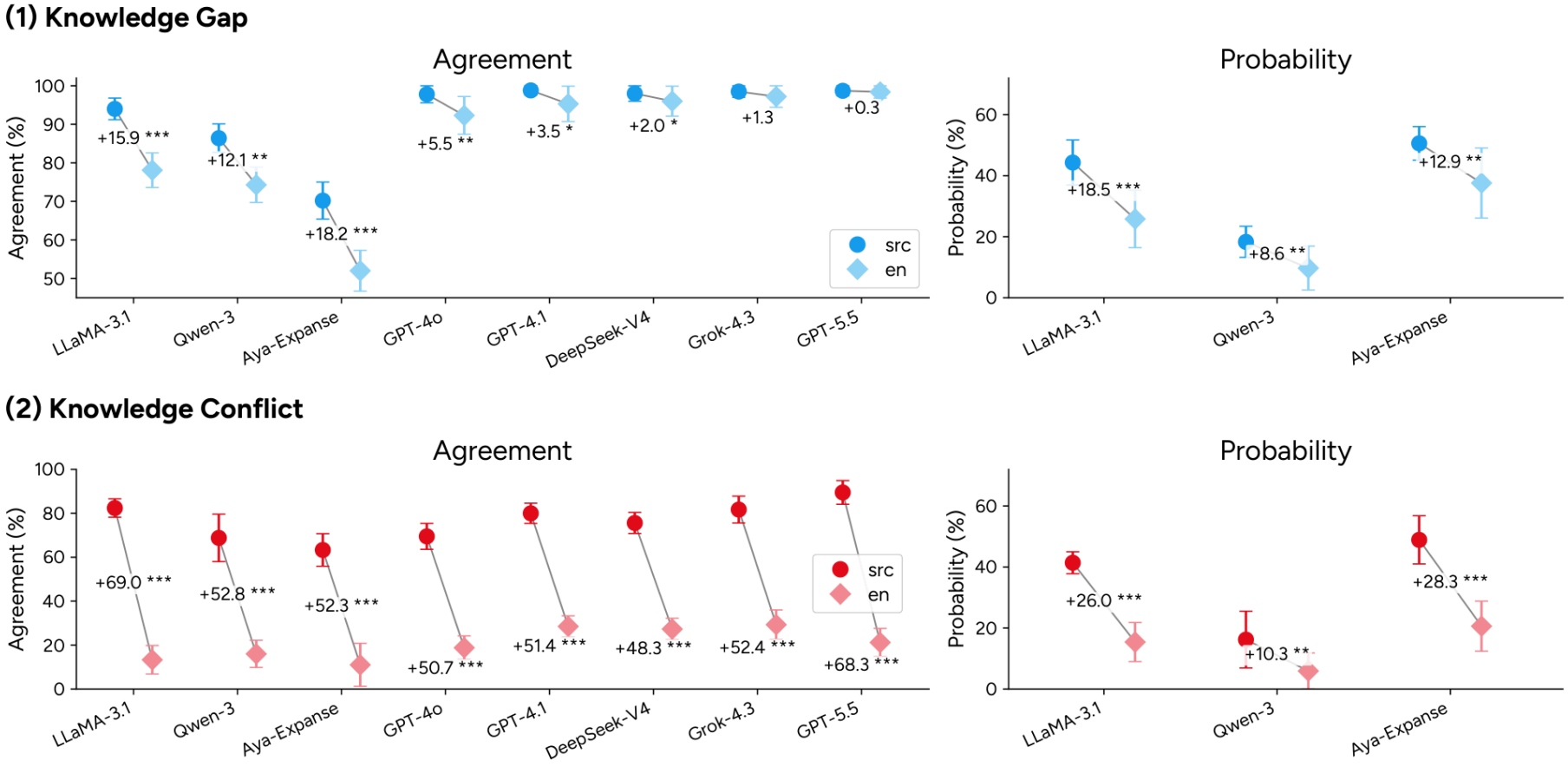
Korean: after 2025.
English: after 2023
(injected by the authors).

Comparison

Same two documents.
Change the query language.

Results

Evidence across models



Agreement: matches the original fact. **Probability:** reference-token likelihood.
src / en: query language · Δ = src – en · 95% bootstrap CIs
Ki et al. (2026), Figure 2 · probabilities reported for open models only

Results

Evidence across models

Conflicts strengthen query-language preference across models.

Agreement gap (pp)	GPT-4o	GPT-4.1	GPT-5.5
Knowledge gap	5.5	3.5	0.3
Knowledge conflict	50.7	51.4	68.3

OpenAI pattern (different samples): narrowing for gaps, widening for conflicts.

Query-language preference is not absolute: across models,
answers can follow the other-language account in either query direction.

Which questions lead models to follow the other-language account?
Are these questions more local to that language's context, or more international?

Ki et al. (2026), Figure 2 · rates vary across models and query languages.

Discussion · 1

Beyond recent factual updates

In this paper: facts come from Wikipedia revisions and are basically Date, Num, Per, Loc, Org, Misc.

§3.1–3.2; Appendix A.5, Table 8

What happens with interpretive knowledge and contested knowledge?

Interpretive knowledge

Explaining why an event happened or what it means.

Contested knowledge

Competing accounts of causes, responsibility, or significance.

Discussion · 2

Documents versus AI search

In this paper: the model receives both Wikipedia documents. The retrieval check adds web documents but always retains the original pair.

§4.1; Appendix B.1, Table 9 · 100 questions, two original + four retrieved documents

Does supplying the documents give them too much weight compared with real-life AI search?

In real search, the system must first decide what to retrieve
and which sources to include.

The supplied pair may not be the evidence a user's query would naturally surface.

Discussion · 3

Where do both phenomena matter?

Language spillover

Ask in Chinese; receive an answer grounded in English sources.

Language preference bias

Query language affects which conflicting account is favored.

In what niche context would these two phenomena have especially important consequences?

Discussion 3 · Example 1

History and sovereignty

Converging “Truths”? The Role of Digital Knowledge Platforms for Information Flows across Language Barriers and History

Studied in Wikipedia: how facts and narratives converge across language editions. This offers a close parallel to the cross-language information differences facing AI.

Facts

Differences in reported troop and casualty numbers.

Narratives

Multilingual text embeddings measure semantic distance.

AI extension

Territorial sovereignty: could query language change how AI presents competing claims?

Converging “Truths”? (2025), pp. 4, 20–24 · Wikipedia evidence; the AI extension is proposed.

Discussion 3 · Example 2

Asylum information

Italy · asylum seekers · Italian, Arabic, and English

Same legal situation, different query language: does AI describe different rights or remedies?

Possible source of conflict

Government, immigration-agency, and NGO guides may differ in translation coverage, update frequency, and delays.

Proposed audit

Hold the case and documents fixed. Vary query language; separate source language, date, and authority.

Measure: answer accuracy, omitted remedies, and advice that could discourage further action.

Verify real version conflicts first; use legal expertise to establish the applicable rule.

EUAA: information provision in Italy · multilingual materials are documented; conflicts remain to be established.

Discussion 3 · Example 3

Mekong water disputes

One low-water event · Chinese, Thai, and English accounts

Does query language change how AI attributes responsibility for unusually low water levels?

Evidence to assemble

Water-agency reports, Mekong River Commission (MRC) analyses, and local accounts of the same event.

Explanations to compare

Rainfall and climate; upstream reservoir operations; downstream infrastructure and other factors.

Citation diversity ≠ explanatory influence.
Does the answer integrate competing accounts, or merely cite them?

Verify competing accounts; assess attribution and uncertainty rather than a single correct explanation.

MRC hydrometeorological monitoring · a starting point for selecting events and evidence.